

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Tin(II) chloride dihydrate**  
 Product Description: **Tin(II) chloride dihydrate**  
 Cat No. : L09080  
 Synonyms Stannous chloride dihydrate  
 CAS No 10025-69-1  
 Molecular Formula Cl<sub>2</sub> Sn . 2 H<sub>2</sub> O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                     |
|------------------------------------------------------|---------------------|
| Substances/mixtures corrosive to metal               | Category 1 (H290)   |
| Acute oral toxicity                                  | Category 4 (H302)   |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 4 (H332)   |
| Skin Corrosion/Irritation                            | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation                    | Category 1 (H318)   |
| Skin Sensitization                                   | Category 1 (H317)   |
| Specific target organ toxicity - (single exposure)   | Category 3 (H335)   |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)   |
| Chronic aquatic toxicity                             | Category 3 (H412)   |

**Label Elements**

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**Signal Word**

**Danger**

## Hazard Statements

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H335 - May cause respiratory irritation  
H373 - May cause damage to organs through prolonged or repeated exposure  
H412 - Harmful to aquatic life with long lasting effects  
H302 + H332 - Harmful if swallowed or if inhaled

## Precautionary Statements

### Prevention

P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P273 - Avoid release to the environment

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P390 - Absorb spillage to prevent material damage

### Storage

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant liner

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                   | CAS No     | Weight % |
|-----------------------------|------------|----------|
| Stannous chloride dihydrate | 10025-69-1 | >95      |
| Stannous chloride           | 7772-99-8  | -        |

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## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

#### Ingestion

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes.

#### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

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## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

### Environmental precautions

Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store in metal containers. Corrosives area. Store under an inert atmosphere.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                   | Malaysia | ACGIH TLV                | OSHA PEL                           |
|-----------------------------|----------|--------------------------|------------------------------------|
| Stannous chloride dihydrate |          | TWA: 2 mg/m <sup>3</sup> | (Vacated) TWA: 2 mg/m <sup>3</sup> |
| Stannous chloride           |          | TWA: 2 mg/m <sup>3</sup> | (Vacated) TWA: 2 mg/m <sup>3</sup> |

| Component                   | European Union | The United Kingdom                                                | Germany |
|-----------------------------|----------------|-------------------------------------------------------------------|---------|
| Stannous chloride dihydrate |                | STEL: 4 mg/m <sup>3</sup> 15 min<br>TWA: 2 mg/m <sup>3</sup> 8 hr |         |
| Stannous chloride           |                | STEL: 4 mg/m <sup>3</sup> 15 min<br>TWA: 2 mg/m <sup>3</sup> 8 hr |         |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

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## Hand Protection Skin and body protection

Protective gloves  
Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Appearance     | Colorless                |
| Physical State | Solid                    |
| Odor           | Odorless                 |
| Odor Threshold | No data available        |
| pH             | No information available |

|                     |                              |
|---------------------|------------------------------|
| Melting Point/Range | 37 - 38 °C / 98.6 - 100.4 °F |
|---------------------|------------------------------|

|                 |                   |
|-----------------|-------------------|
| Softening Point | No data available |
|-----------------|-------------------|

|                     |                    |
|---------------------|--------------------|
| Boiling Point/Range | 652 °C / 1205.6 °F |
|---------------------|--------------------|

|             |                          |
|-------------|--------------------------|
| Flash Point | No information available |
|-------------|--------------------------|

@ 760 mmHg

**Method -** No information available

|                  |                |
|------------------|----------------|
| Evaporation Rate | Not applicable |
|------------------|----------------|

|                          |                          |
|--------------------------|--------------------------|
| Flammability (solid,gas) | No information available |
|--------------------------|--------------------------|

|                  |                   |
|------------------|-------------------|
| Explosion Limits | No data available |
|------------------|-------------------|

Solid

|                |                   |
|----------------|-------------------|
| Vapor Pressure | No data available |
|----------------|-------------------|

|               |                |
|---------------|----------------|
| Vapor Density | Not applicable |
|---------------|----------------|

|                            |                   |
|----------------------------|-------------------|
| Specific Gravity / Density | No data available |
|----------------------------|-------------------|

|              |                   |
|--------------|-------------------|
| Bulk Density | No data available |
|--------------|-------------------|

|                  |                 |
|------------------|-----------------|
| Water Solubility | 1187 g/l (20°C) |
|------------------|-----------------|

|                              |                          |
|------------------------------|--------------------------|
| Solubility in other solvents | No information available |
|------------------------------|--------------------------|

Solid

Partition Coefficient (n-octanol/water)

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|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | No data available        |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |

|                   |                                         |
|-------------------|-----------------------------------------|
| Molecular Formula | Cl <sub>2</sub> Sn . 2 H <sub>2</sub> O |
| Molecular Weight  | 225.63                                  |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available. Strong reducing agent. Fire and explosion risk in contact with oxidizing agents.

### Chemical Stability

Strong reducing agent. Fire and explosion risk in contact with oxidizing agents. Moisture sensitive. Air sensitive.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### Conditions to Avoid

Avoid dust formation. Incompatible products. Excess heat. Exposure to moist air or water.

### Incompatible Materials

Strong oxidizing agents. Peroxides. Alkali metals. . Nitrates. Ethylene oxide.

### Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Category 4                                                       |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Category 4                                                       |

| Component         | LD50 Oral                 | LD50 Dermal | LC50 Inhalation                  |
|-------------------|---------------------------|-------------|----------------------------------|
| Stannous chloride | LD50 = 1910 mg/kg ( Rat ) | -           | LC50 = 2mg/l (4h) rat (OECD 436) |

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(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

Skin

Based on available data, the classification criteria are not met

Category 1

No information available

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

| Component                            | Test method                                   | Test species          | Study result |
|--------------------------------------|-----------------------------------------------|-----------------------|--------------|
| Stannous chloride<br>7772-99-8 ( - ) | OECD Test Guideline 476<br>Gene cell mutation | in vitro<br>Mammalian | negative     |

Mutagenic effects have occurred in humans

(f) carcinogenicity;

Based on available data, the classification criteria are not met

| Component                            | Test method             | Test species / Duration | Study result |
|--------------------------------------|-------------------------|-------------------------|--------------|
| Stannous chloride<br>7772-99-8 ( - ) | OECD Test Guideline 451 | Rat<br>mouse<br>2 years | negative     |

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

| Component                            | Test method                                | Test species / Duration | Study result                    |
|--------------------------------------|--------------------------------------------|-------------------------|---------------------------------|
| Stannous chloride<br>7772-99-8 ( - ) | OECD Test Guideline similar to<br>OECD 416 | rabbit<br>15 days       | NOAEL =<br>41.5<br>mg/kg bw/day |

(h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system.

(i) STOT-repeated exposure;

Category 2

Target Organs

Respiratory system, Eyes, Skin, Kidney, Blood, spleen.

(j) aspiration hazard;

Not applicable

Solid

Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

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## Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component         | Freshwater Fish | Water Flea           | Freshwater Algae | Microtox |
|-------------------|-----------------|----------------------|------------------|----------|
| Stannous chloride |                 | EC50 = 19.5 mg/L/48h |                  |          |

## Persistence and degradability

### **Persistence**

Soluble in water, Persistence is unlikely, based on information available.

### **Degradability**

Not relevant for inorganic substances.

### **Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

## Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Should not be released into the environment Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment Large amounts will affect pH and harm aquatic organisms

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### **UN-No**

UN3260

#### **Hazard Class**

8

#### **Packing Group**

III

#### **Proper Shipping Name**

CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. Tin (II) chloride dihydrate

### Road and Rail Transport

#### **UN-No**

UN3260

#### **Hazard Class**

8

#### **Packing Group**

III

#### **Proper Shipping Name**

CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. Tin (II) chloride dihydrate

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## IATA

UN-No UN3260  
Hazard Class 8  
Packing Group III  
Proper Shipping Name CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. Tin (II) chloride dihydrate

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                   | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Stannous chloride dihydrate | -         | -    | -   | X     | X    | X    | X     | X    | -        |
| Stannous chloride           | 231-868-0 | X    | X   | X     | X    | X    | X     | X    | KE-33845 |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

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|                  |                                             |
|------------------|---------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department |
| Revision Date    | 25-Mar-2025                                 |
| Revision Summary | SDS sections updated.                       |

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**